Project title: Intelligent Traffic Forecasting Using LSTGAN

- **Introduction:**
  - Predict future traffic speeds across $N = 325$ sensors using a **Long-Term Spatio-Temporal Graph Attention Network (LSTGAN)**
  - City road network abstracted as spatio-temporal graph $\mathcal{G}(\mathcal{V}, \mathcal{E}, \mathrm{A})$; $\mathrm{A} \in \mathbb{R}^{N \times N}$ adjacency matrix
  - Goal: learn mapping $\mathrm{X}_{t\text{-}P\text{-}1:t} \xrightarrow{f(\cdot)} \mathrm{X}_{t+1:t+H}$ (P past steps $\rightarrow$ predict next $H = 12$ steps; evaluated at 15, 30, 60 min)
- **Motivation:**
  - **Spatial ("Where")**: crash on off-ramp backs up highway; distant school zones share 3 PM spikes — require *both* local and global spatial modeling
  - **Temporal ("When")**: immediate momentum (last 60 min) *plus* deep cycles (daily rush hour, weekly rhythm)
  - Goal: proactive signal control, dynamic routing, faster emergency dispatch, reduced urban congestion

Existing work / Literature review:

- **DCRNN** (Li et al. 2018) — Diffusion conv on directed graph; replaces GRU matrix multiply with random walks Diffusion conv: $\sum_k (\alpha_k (\mathbf{D}_O^{-1}\mathbf{A})^k + \beta_k (\mathbf{D}_I^{-1}\mathbf{A}^\top)^k)\mathbf{X}$; bidirectional flow modeling
- **Graph WaveNet** (Wu et al. 2019) — Self-adaptive adjacency matrix + dilated causal 1D convolutions Learns $\tilde{A}$ without pre-defined graph; WaveNet-style temporal conv; benchmarked on METR-LA & PEMS-BAY
- **ASTGCN** (Guo et al. 2019) — Attention Based Spatial-Temporal Graph Convolutional Networks Introduced segmenting data into recent, daily, and weekly periods; relies on pre-defined local graph & lacks dynamic global spatial attention.

- **Gaps in the existing solutions/work:**

  G1. DCRNN & GWNet model only **local spatial neighbors** — no global attention to discover semantically similar but physically **distant** sensors (e.g. two school zones)

  G2. Existing models use a **single recent window** — no multi-scale temporal view incorporating previous-day and previous-week cycles simultaneously

  G3. No explicit **time encoding** ($\mathbf{V}^d + \mathbf{V}^t$) — models cannot inherently distinguish Tuesday 8 AM rush from Sunday 8 AM calm

Work Done till date:

- **Details of proposed work/solutions:**

  Solving G1 (local-only spatial): LSTGAN fuses **Local GCN** ($X_{GCN} = \sum_{k=0}^{K-1} \Theta^{(k)} Z^{(k)}$, $K = 5$) with **Global Spatial Attention** ($\alpha_{ij} = \mathrm{softmax}(q_i \cdot k_j^{\top} / \sqrt{d_k})$) for physical/semantic correlations.

  Solving G2 (single-scale temporal): Multi-scale 1D Conv encoder processes recent, previous-day, and previous-week windows; compressed $T' = \lfloor (T-k)/s \rfloor + 1$. Temporal attention decoder weighs all evidence dynamically.

  Solving G3 (no time encoding): Inject timestamps $\mathrm{STE} = \mathrm{Norm}(\mathrm{ST}_e + V^t + V^d)$; day-of-week V^d and time-of-day V^t vectors fused into spatial features.

- **Result achieved (if any):**
  - **Data Pipeline & Temporal Features:** Processed 6-mo PeMS-BAY data (~52k steps); engineered time embeddings (weekdays, rush hours) for LSTGAN's V^d and V^t.
  - **Graph Construction:** Handled missing values (bidirectional fill, Z-score); built and validated a 2,694-edge Gaussian spatial graph.
  - **Advanced EDA Dashboards:** Generated interactive tools (sensor maps, speed heatmaps, daily rhythms) to guarantee data quality before training.